SOFTWARE ENGINEERING

Lab 6: CODE AND BRANCH ANALYSIS

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CLASS: AIML SEC 'H' 5TH SEMESTER

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ASSIGNED PROBLEM: Gym Membership Management System (Problem 21)

Initial Code Coverage

```
C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>python -m pytest -v
============================================= test session starts =============================================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\Ananya\AppData\Local\Programs\Python\Python314\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6
plugins: anyio-4.15.1, platformdirs-4.12.1
collected 2 items

test_gym_membership.py::test_monthly_plan PASSED                                                      [ 50%]
test_gym_membership.py::test_can_enter PASSED                                                         [100%]

============================================== 2 passed in 0.11s ==============================================

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>python -m coverage run -m pytest
============================================= test session starts =============================================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6
plugins: anyio-4.15.1, platformdirs-4.12.1
collected 2 items

test_gym_membership.py ..                                                                             [100%]

============================================== 2 passed in 0.03s ==============================================

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>python -m coverage report -m
Name                    Stmts   Miss  Cover   Missing
-----------------------------------------------------
gym_membership.py          35     20    43%   6, 8, 11, 13, 15-17, 23, 25, 27, 32-41
test_gym_membership.py      6      0   100%
-----------------------------------------------------
TOTAL                      41     20    51%

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>
```

Initial Branch Coverage

```
C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>python -m coverage erase

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>python -m coverage run --branch -m pytest
============================================= test session starts =============================================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6
plugins: anyio-4.15.1, platformdirs-4.12.1
collected 2 items

test_gym_membership.py ..                                                                             [100%]

============================================== 2 passed in 0.04s ==============================================

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>python -m coverage report -m
Name                    Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------
gym_membership.py          35     20     26      8    38%   6, 8, 11, 13, 15-17, 23, 25, 27, 32-41
test_gym_membership.py      6      0      0      0   100%
-----------------------------------------------------------------
TOTAL                      41     20     26      8    43%

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>
```

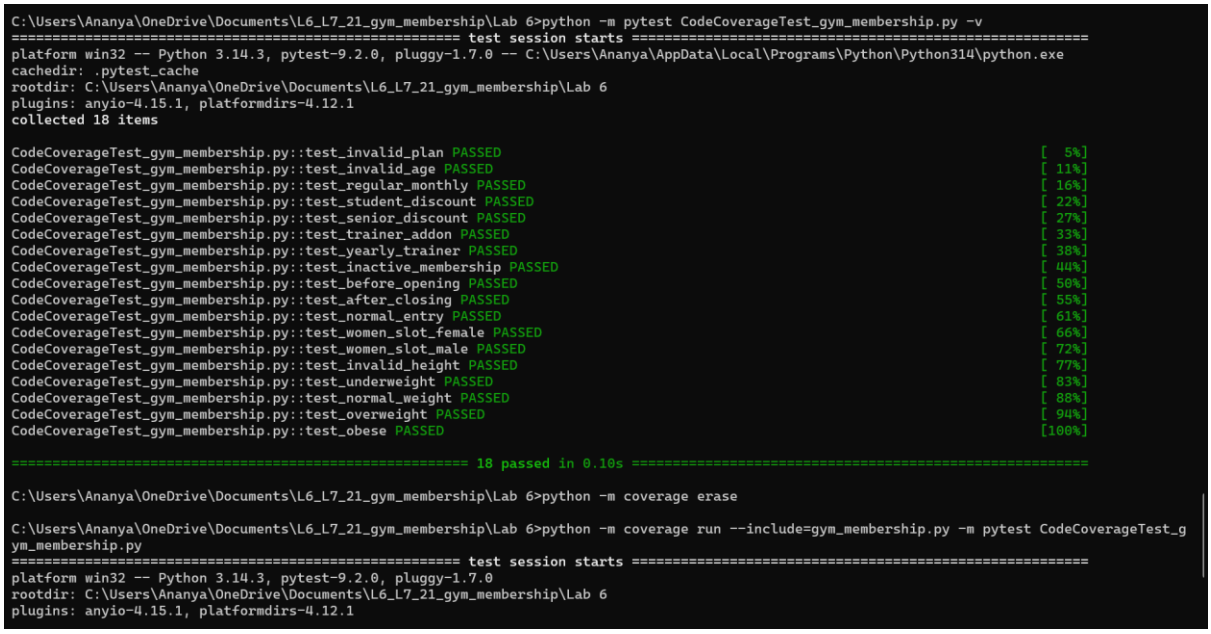
Code Coverage Analysis

The file CodeCoverageTest_gym_membership.py contains 18 test cases covering the application functions, including membership cost calculation, gym entry eligibility and BMI classification.

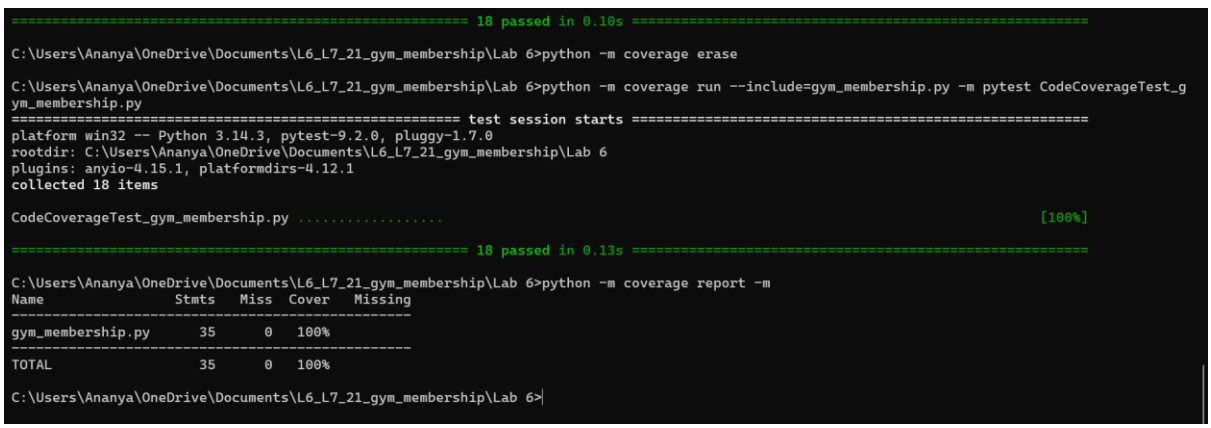
After executing the test cases, all 35 executable statements in gym_membership.py were covered.

- Total statements: 35
- Statements missed: 0
- Code coverage achieved: 100%

Code Coverage Test Execution



Final Code Coverage Report



Branch Coverage Analysis

The file BranchCoverageTest_gym_membership.py contains 21 test cases designed to exercise the conditional branches in the application.

The tests cover different membership conditions, age categories, student discounts, trainer options, gym entry restrictions and BMI classifications.

The final coverage report showed:

- Total branches: 26
- Branches missed: 0
- Partial branches: 0
- Branch coverage achieved: 100%

Branch Coverage Test Execution

```
C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>python -m pytest BranchCoverageTest_gym_membership.py -v
============================================================ test session starts =============================================================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\Ananya\AppData\Local\Programs\Python\Python314\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6
plugins: anyio-4.15.1, platformdirs-4.12.1
collected 21 items

BranchCoverageTest_gym_membership.py::test_unknown_plan PASSED                                                                      [  4%]
BranchCoverageTest_gym_membership.py::test_age_below_14 PASSED                                                                      [  9%]
BranchCoverageTest_gym_membership.py::test_senior_discount PASSED                                                                   [ 14%]
BranchCoverageTest_gym_membership.py::test_student_discount PASSED                                                                  [ 19%]
BranchCoverageTest_gym_membership.py::test_regular_non_student PASSED                                                               [ 23%]
BranchCoverageTest_gym_membership.py::test_trainer_monthly PASSED                                                                   [ 28%]
BranchCoverageTest_gym_membership.py::test_trainer_yearly PASSED                                                                    [ 33%]
BranchCoverageTest_gym_membership.py::test_senior_with_trainer PASSED                                                               [ 38%]
BranchCoverageTest_gym_membership.py::test_inactive_plan PASSED                                                                     [ 42%]
BranchCoverageTest_gym_membership.py::test_before_opening PASSED                                                                    [ 47%]
BranchCoverageTest_gym_membership.py::test_opening_boundary PASSED                                                                  [ 52%]
BranchCoverageTest_gym_membership.py::test_closing_boundary PASSED                                                                  [ 57%]
BranchCoverageTest_gym_membership.py::test_after_closing PASSED                                                                     [ 61%]
BranchCoverageTest_gym_membership.py::test_normal_slot PASSED                                                                       [ 66%]
BranchCoverageTest_gym_membership.py::test_women_slot_female PASSED                                                                 [ 71%]
BranchCoverageTest_gym_membership.py::test_women_slot_male PASSED                                                                   [ 76%]
BranchCoverageTest_gym_membership.py::test_invalid_height PASSED                                                                    [ 80%]
BranchCoverageTest_gym_membership.py::test_underweight PASSED                                                                       [ 85%]
BranchCoverageTest_gym_membership.py::test_normal_bmi PASSED                                                                        [ 90%]
BranchCoverageTest_gym_membership.py::test_overweight_bmi PASSED                                                                    [ 95%]
BranchCoverageTest_gym_membership.py::test_obese_bmi PASSED                                                                         [100%]

============================================================ 21 passed in 0.06s =============================================================

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>
```

Final Branch Coverage Report

```
C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>python -m coverage erase

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>python -m coverage run --branch --include=gym_membership.py -m pytest BranchCov
erageTest_gym_membership.py
============================================================ test session starts =============================================================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6
plugins: anyio-4.15.1, platformdirs-4.12.1
collected 21 items

BranchCoverageTest_gym_membership.py .....................                                                                          [100%]

============================================================ 21 passed in 0.05s =============================================================

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>python -m coverage report -m
Name                  Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------
gym_membership.py        35      0     26      0   100%
-----------------------------------------------------------------
TOTAL                    35      0     26      0   100%

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 6>
```

Overall Report

The experiment successfully achieved both required coverage targets. The final results are:

Metric	Initial Coverage	Final Coverage
Code Coverage	43%	100%
Branch Coverage	38%	100%

A total of 39 additional test cases were implemented across two separate test files. Both test suites passed successfully.